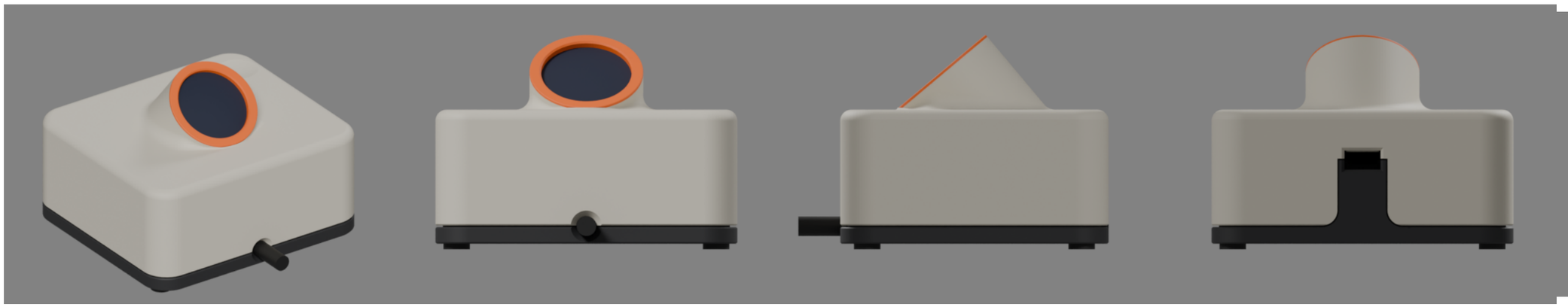
PSU tester case - concept I9 (one straight 40 deg pipe, rounded root, no trailing tunnel; I8 ring ribs) - review CAD (exact B-rep) | 2026-10-21 | all dims mm

Isometric

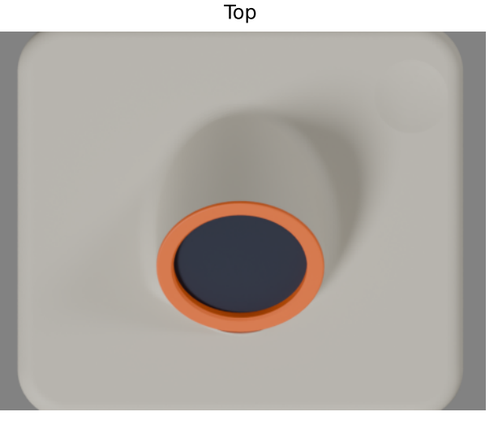


Front (user side)

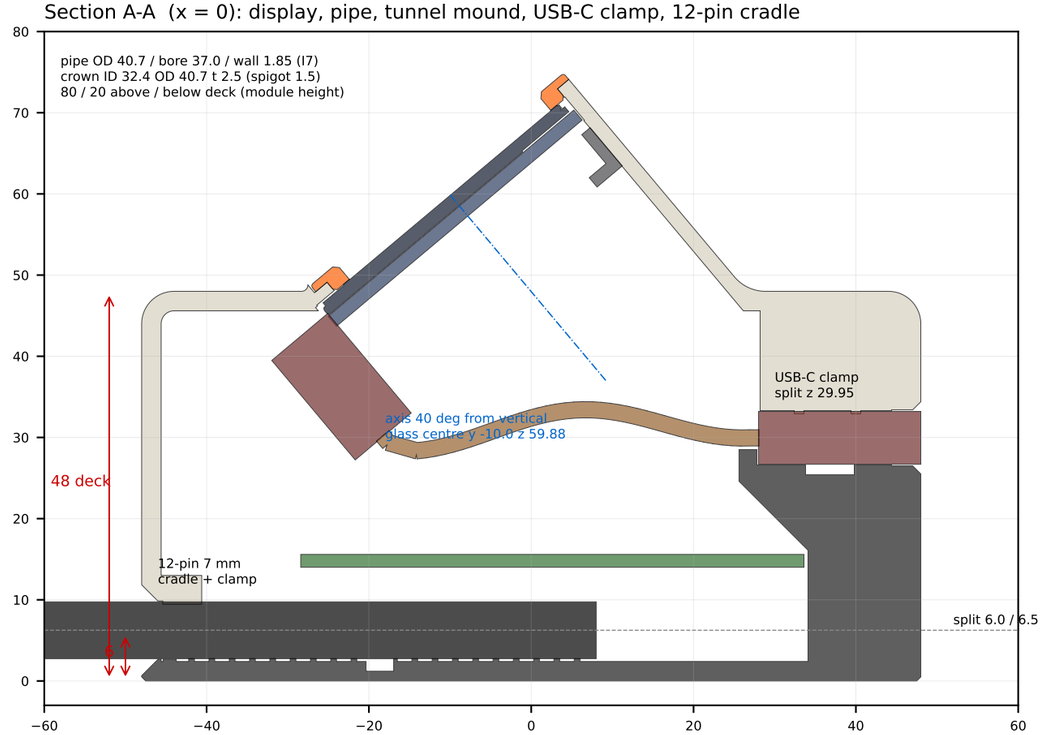
Side (from right)

Rear (USB-C)

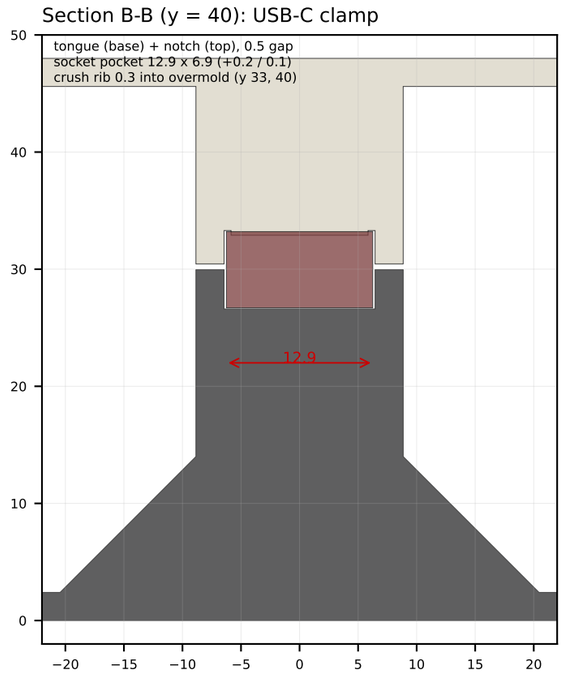
Top



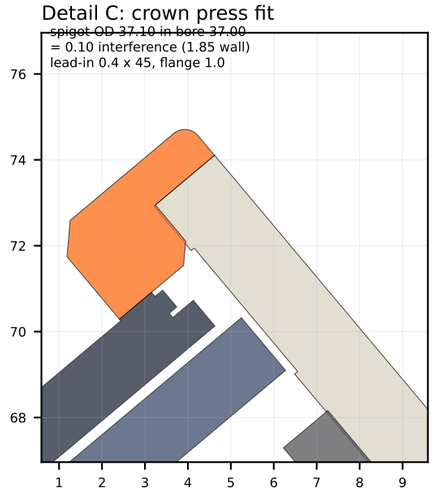
Section A-A (x = 0): display, pipe, tunnel mound, USB-C clamp, 12-pin cradle



Section B-B (y = 40): USB-C clamp



Detail C: crown press fit



Section colours



KEY DIMENSIONS (I7 changes marked *)
Footprint 108 x 96, R12 plan corners; deck z 48 (R4 edge); pipe top z 74.1
Base 0-6 (black band) + 0.5 shadow gap; top from 6.5; wall 2.4; floor 2.4
*Bore d37.00 (was 36.70), pipe OD 40.7 -> wall 1.85; module seat/tab cut = real PCB+LCM outline + 0.2
*Display = Waveshare vendor STEP: PCB OD 36.5 (+0.13 u offset), 39.5 tall, trapezoid USB tab 12.8 at the bottom; LCM 35.6; 40 deg tilt, glass centre (0, -10, 59.88)
Crown ID 32.4 / OD 40.7, spigot into the bore; *retainer sleeve OD 37.10 / ID 34.4
*PCB = Rev D2 KiCad model (tester-d2.wrl) 86x62x1.6 on 4 standoffs d6 (z 14), pilot d2.0 x 12; holes (+-40, PCB_Y+-28) match
4x shell screws from below at (+/-46.41, +/-38.36): d3.5 thru, c'bore d5.5 x 8.0 (grommet), floor 2.5
Top bosses d7.2 from z 11.0, pilot d2.0 x 10.5; deck solid (no vents)
*I7b: 12-pin cradle 7.3 wide, ONE tie at the trench middle y -18.65: 3.0 x 1.6 wall slots at mid-height (z 4.4-6.0), inner grooves + blind bed pocket (floor 1.2 left), blind head pocket 6.0 x 5.8 right of +x wall; underside flat
*Jacket to y 8.0; 12 conductors d1.4 fan to J1-J3 (U-arc under the board at z 10.95, R3.5 up in the front zone)
USB-C socket centre z 29.95, tongue 16.9 wide, R6 flares

PARTS LIST

I9_base	Bambu PLA Matte Charcoal	prints flat, no supports
I9_top	PLA Matte Ivory White*	upright on split edge, tree supports inside only
I9_crown	PLA Basic Orange* (#FF6A13)	flange down, no supports
I9_retainer	PLA Matte Charcoal (hidden)	sleeve, flange down, no supports

8x M2.5 x 12 plastic thread-forming flat(pan)-head screw (4 shell + 4 PCB); 4x rubber grommet foot (hip 5.5 x 8 ASSUMED)
1x Hirose HR10A-10P-12S(73) female plug on 7 mm 12-core cable (12 x 0.34 mm2 ASSUMED)
12x insulated ferrule, 8.0 mm (0.314") pin d1.1, green collar d2.6 (type/size to confirm vs conductor gauge)
1x lead JST-XHP-5 (J4) -> 5 x 28 AWG -> 5 single round 1-pin housings d1.2 x 12 on H2.1/.3/.4/.5/.7 (0.64 sq pins)
1x zip tie 2.5 mm at the trench middle (I7b; through both wall slots, under the jacket); no heat-set inserts
Waveshare ESP32-S3-LCD-1.28; YANBORONSN USB4 90 deg ext. 51 mm
* colour names from the Bambu catalogue - verify spool names

VALIDATION
All 4 print solids valid, closed after STEP re-import; vol cm3 base 43.27, top 68.855, crown 0.829, retainer 1.156
Intended: top x crown 6.547, top x retainer 28.647 (press fits), top x female 8.424 (crush ribs), 12-pin clamp
*Real module: PCB x top 0.0 (I6 9.19), LCM x top 0.0 (I6 4.57) mm3; gaps PCB-top 0.2, LCM-top 0.2
*Real PCBA (fcl mesh): top-board 2.258, top-components >= 10.443, base-SMD 2.258
*Conductors: to THT terminal pins/bodies 6.995, base 0.084, top 6.993, board 1.722; collars-board 1.044
*H2 housings d1.2: gap between 0.07, to sleeve 0.985, to top 2.584; leads-top 2.784
USB: top-male plug 1.0, top-flat 2.2, base-flat 0.43; flip check: 180 deg -> 322.262 mm3 into the top (not buildable)
Overhangs: base support-free (tie slots = 3.0 bridges); crown/retainer none; top: tree supports inside only

OPEN ITEMS
1. USB 'flip': 180 deg about the plug axis puts the 90 deg overmold 8.5 mm in front of the glass (through the deck) -> kept as I6; confirm what 'flip' means
2. Conductor -> terminal mapping, gauge (0.34 mm2 assumed, d1.4) and ferrule type (0.314" pin, green collar) to confirm
3. H2 pin numbering (pin 1 at +u, odd pins outer row) unverified; d1.2 housings leave 0.07 between neighbours at 1.27 pitch; 0.64 pins in a 1.27 socket: check
4. Module: Petter's 36.4 vs Waveshare/STEP 36.5; art 31.3 vs LCD AA 32.4 - crown ID 32.4 kept; bore 37.0 gives 0.11 at the PCB top point
5. USB overmold dims are ESTIMATES - caliper, then tune POCKET
6. Crown + retainer press fit 0.10 dia interference: test print, tune 0.10-0.15
7. Grommet geometry (hip 8.0) assumed; colour/filament names unverified
8. Faceted PCBA/vendor bodies make the assembly STEP large (~160 MB); per-part STEPs are small exact B-rep